

Delay Selection by Spike-Timing-Dependent Plasticity Shapes Efficient Networks for Signal Transmission

Ali Ghadiri,¹ Saeed Taghavi,² Hedyeh Rezaei,¹  Ad Aertsen,³ Arvind Kumar,^{4,5} and Alireza Valizadeh^{1,2}

¹Department of Physics, Institute for Advanced Studies in Basic Sciences (IASBS), Zanjan 45195-1159, Iran, ²The Zapata-Briceño Institute of Neuroscience, Madrid 28015, Spain, ³Faculty of Biology, Bernstein Center Freiburg, University of Freiburg, Freiburg 79104, Germany, ⁴Department of Computational Science and Technology, School of Computer Science and Communication, KTH Royal Institute of Technology, Stockholm 11428, Sweden, and ⁵Science for Life Laboratory, Stockholm 17165, Sweden

Information processing in the brain relies on efficient communication between different brain regions. Brain oscillations can control signal transmission in brain networks by modulating the timing and excitability of sender and receiver areas. For effective transmission, signals should arrive at target areas when their excitability is maximal. For reciprocally connected neural populations, this mechanism works if the transmission delay matches the period of their evoked oscillation. However, the mechanisms underlying such development of the connections with matched delays remain elusive. While transmission delays in brain networks change during development, the process by which delays are tuned for efficient transmission is unknown. Here, we demonstrate that the well-known Hebbian learning rule can provide a mechanism for selecting connections with delays that match the period of network oscillations. We consider a reciprocally connected bi-layer network of excitatory and inhibitory neurons that generate network-level oscillations spontaneously or in response to external stimuli. When exposed to spike-timing-dependent plasticity, the network self-organizes to potentiate connections with delays matching the oscillation period, while depressing those with non-matching delays. Our findings shed light on how transmission delays may evolve during learning and development to optimize the organization of brain networks for efficient signal transmission.

Key words: learning; resonance frequency; self-organization; signal transmission; STDP rule; synaptic delay

Significance Statement

Efficient communication between brain regions requires that signals arrive when the target population is most excitable, which depends on the matching of transmission delays with the ongoing rhythm in the network. How such matched delays arise has remained unclear. Here we argue that they need not be set by any mechanism acting directly on delays. We show that ordinary Hebbian spike-timing-dependent plasticity can select connections with suitable delays depending on the state of local dynamics of the two brain regions. Our results specifically link synaptic plasticity to network rhythms and suggest how inter-areal communication delays may be tuned during learning and development. These results imply that even inter-areal physical connectivity in the brain may be contingent on brain activity.

Introduction

Controllable and flexible communication between different regions along the processing hierarchy and between parallel processing pathways (Kaas, 1997; Fries, 2005; Akam and Kullmann, 2014;

Kohn et al., 2020) is essential for brain function. Collective oscillations of neuronal populations have been proposed as a mechanism for efficient communication between neurons at different spatial scales (Gray et al., 1989; Brovelli et al., 2004; Fries, 2005, 2009; Bastos et al., 2015; Fries, 2015; Bonnefond et al., 2017; Wang and Tang, 2019). From a theoretical viewpoint, coordinated neuronal activity engaging population oscillations establishes a mechanism for generating stronger signals (Pouille and Scanziani, 2001; Womelsdorf and Fries, 2007), while the change in excitability of receiver regions over each oscillation cycle can determine the effectiveness of incoming signals (Aertsen et al., 1996; Fries et al., 2007; Womelsdorf and Fries, 2007; Akam and Kullmann, 2014). Oscillations can arise from a resonance-like response to external stimuli or due to the dynamical state of the brain (Neuenschwander and Varela, 1993; Haenschel et al., 2009).

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Correspondence should be addressed to Alireza Valizadeh at valizadeh@gmail.com or Arvind Kumar at arvinkumar@kth.se.

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Oscillation-based communication between connected brain networks requires synchronization at a fixed latency. However, such connections always carry transmission delays that depend on inter-regional distance, axonal conduction velocity, and fiber myelination, all of which span a broad range (Kusano, 1966; Kress et al., 2008; Pérez et al., 2011; Ziaemehr et al., 2020; Pariz et al., 2021). Experimental and anatomical studies suggest that cortico-cortical delays can vary from a few milliseconds for nearby regions to several tens of milliseconds for long-range projections, with reported values typically ranging from 1–20 ms and up to 50 ms, depending on pathway length and axonal properties (Girard et al., 2001; Salami et al., 2003; Swadlow and Waxman, 2012; Caminiti et al., 2013). Key questions are how the brain coordinates spike-timing, despite this wide range of delays (Knoblauch and Palm, 2002; Pérez et al., 2011) and whether regulatory processes adjust the delays within and between cortical areas (Sugihara et al., 1993; Pelletier and Paré, 2002; Salami et al., 2003; Lang and Rosenbluth, 2003; Chomiak et al., 2008; Seidl, 2014). Theoretical studies have proposed specific rules for myelination changes in axons of the central nervous system that might tune the spike arrival time by regulating the transmission delays (Pajevic et al., 2023). However, while there is ample evidence for changes in myelination during learning (Fields, 2008), the rules governing this process remain obscure. It remains unclear whether myelination is a dedicated learning process or a homeostatic mechanism regulating coarse-grained activity, and whether it shapes structure mainly during development or acts throughout life, like plasticity (Williamson and Lyons, 2018). Spike-timing-dependent plasticity (STDP) can also form a mechanism to fine-tune connection delays by selectively strengthening or weakening connections of specific delays (Gerstner et al., 1996; Kerr et al., 2013).

In this study we propose that the well-known Hebbian STDP rule can shape the distribution of transmission delays for efficient communication on its own, without a distinct rule acting directly on the delays. To demonstrate this, we investigated a bi-directional two layer network of spiking neurons whose inter-layer synapses were plastic and updated by STDP, asking how plasticity shapes inter-layer connectivity strengths and delays through the network's resonance/oscillation property. Starting from a wide distribution of delays, we found that STDP consistently potentiated connections whose forward and backward delays summed to the oscillation period of the two networks. As shown previously, such a match between two seemingly distinct time scales is crucial for efficient signal transmission along layered networks (Hahn et al., 2014; Rezaei et al., 2020). Thus, through STDP the network self-organizes to a consistent structure in which the delays match the oscillation period, whether the oscillations are evoked or spontaneous. This reveals a deep interplay between synaptic plasticity and network dynamics, relating two key properties of the network—resonance/oscillation frequency and the plasticity rule—as it tunes its temporal structure to optimize transmission.

Methods

Neuron and synapse model

The neuron model simulates a spiking neuron by employing integrate-and-fire dynamics (Eq. 1) using conductance-based synapses:

$$C_m \frac{dV_m}{dt} = -G_{\text{leak}}(V_m(t) - V_{\text{reset}}) + I_{\text{syn}}(t), \quad (1)$$

where V_m denotes the membrane potential, C_m the membrane

capacitance, G_{leak} the membrane leak conductance, V_{reset} the reset potential after initiating a spike, and I_{syn} the total synaptic input current.

When the membrane voltage reached the threshold for spiking ($V_{\text{th}} = -54$ mV), a spike was emitted, and the membrane potential was reset and clamped to the resting membrane potential ($V_{\text{reset}} = -70$ mV) for a refractory period of $\tau_{\text{ref}} = 2$ ms.

To avoid a transient network synchrony at the beginning of the simulation, initial membrane potentials of neurons were drawn from a uniform distribution (low: -70 mV; high: -60 mV). Neuron model parameters are listed in Table 1.

Synaptic inputs were introduced through a transient modification of the synaptic conductance G_{syn} :

$$I_{\text{syn}}(t) = G_{\text{syn}}(t)(V_m(t) - E_{\text{syn}}), \quad (2)$$

where E_{syn} is the synaptic reversal potential. Conductance changes were modeled as single exponential functions:

$$G_{\text{syn}}(t) = G_{\text{syn}} \exp\left(\frac{-(t - t_s)}{\tau_{\text{syn}}}\right), \quad (3)$$

where G_{syn} denotes the connection strength, and τ_{syn} is the synaptic time constant. This function is normalized to ensure that a spike event with a weight of 1.0 corresponds to a peak conductance of 1 nS. Synapse model parameters are given in Table 1. Synaptic transmission delays were set to 1 ms (excitation) and 2 ms (inhibition) for intra-layer connections, whereas inter-layer feedforward (FF) and feedback (FB) transmission delays were drawn from normal distributions, the parameters of which (mean and standard deviation) were varied as mentioned in each figure caption.

Inter-layer connections were among excitatory neurons in the two successive layer networks. In the case of static connections, inter-layer connections (for FF connections only), the synapses were as strong and as probable as within-layer excitatory-to-excitatory connections. For plastic inter-layer connections (involving FF or FB connections as mentioned in each figure caption), though, synapse weights were subjected to a plasticity rule.

Synaptic weights of plastic inter-layer connections varied according to a temporally asymmetric Hebbian STDP rule as follows:

$$\Delta w = \begin{cases} -\lambda \alpha e^{-\frac{|\Delta t|}{\tau_-}} & \Delta t < 0, \\ \lambda e^{-\frac{|\Delta t|}{\tau_+}} & \Delta t \geq 0 \end{cases}. \quad (4)$$

Table 1. Delay selection by spike-timing-dependent plasticity shapes efficient networks for signal transmission

Name	Value	Description	Name	Value	Description
Neuron model parameters					
C_m	250 pF	Membrane capacitance	G_{leak}	16.67 nS	Membrane leak
V_{th}	-54 mV	Firing threshold	V_{reset}	-70 mV	Reset potential
τ_{ref}	2 ms	Refractory period			
Synapse model parameters					
τ_{exc}	3 ms	Excitatory decay	τ_{inh}	3 ms	Inhibitory decay
$E_{\text{syn}}^{\text{exc}}$	0 mV	Excitatory reversal	$E_{\text{syn}}^{\text{inh}}$	-80 mV	Inhibitory reversal
J_{ee}	0.33 nS	E→E	J_{ei}	5.5 nS	E→I
J_{ie}	-6.2 nS	I→E	J_{ii}	-15 nS	I→I
J_{pe}	0.25 nS	Poisson→E	J_{pi}	0.4 nS	Poisson→I
J_{pp}	0.6 nS	Pulse packet→P	d_{E}	1 ms	Excitatory delay
d_{I}	2 ms	Inhibitory delay			
STDP parameters					
λ	0.005	Learning rate	α	0.9	Depress./potent. scaling
τ_+	15.9 ms	Potentiation	τ_-	19.3 ms	Depression
Network model parameters					
N_{exc}	200	Excitatory neurons	N_{inh}	50	Inhibitory neurons
N_{proj}	70	Projection neurons	ϵ	0.2	Within-layer prob.
ϵ_{pp}	1	Inter-layer prob.			

Here, $\Delta t = t_{\text{post}} - t_{\text{pre}}$ denotes the time difference between postsynaptic and presynaptic spikes. The parameters τ_+ and τ_- are the time constants of the potentiation and depression windows, respectively, λ is the learning rate, and α is a scaling factor that controls the relative strength of synaptic depression to synaptic potentiation.

The specific values of these parameters are summarized in Table 1. In this study, we considered the case in which all spike pairs contribute to synaptic weight updates. Depending on the relative timing of the spike pairs, the plasticity rule led to potentiation or depression, respectively. The STDP rule applied in this manuscript was weight-independent, by which a bimodal distribution for final weights was achieved. This, in fact, was a positive mark, because two classes of synapses were distinguishable: very weak and very strong connections. Therefore, synapses that were stronger than the mean initial weight were considered potentiated and contributed to the final distribution of delays. Otherwise, they were considered depressed and played a less or no important role in the final distribution and, hence, in signal propagation. However, as described, a hard upper boundary was required to avoid instabilities in the network dynamics, due to over-potentiation of inter-layer connections.

Instead, a weight-dependent STDP rule, in general, takes care of this kind of instability by incorporating the current weights in weight updates. However, it hardly takes into account clearly differentiated depressed and potentiated synapses. Hence, we applied a weight-independent STDP rule with a hard upper boundary (W_{max}) in our layered networks to study the final distribution of potentiated inter-layer delays and investigate their role in signal propagation across the network.

Network connectivity

The network consisted of two layers, each comprising 200 excitatory and 50 inhibitory neurons (Fig. 1a). Within a layer, 70 neurons projected to the other layer. Random connectivity within the layers was established with a fixed intra-layer connection probability of 0.2 for all types of connections (EE, EI, IE, and II). The intra-layer synaptic weights were kept constant at the values specified in Table 1.

Initially, the study focused on the asynchronous-irregular (AI, Brunel, 2000) regime, characterized by network-level resonance properties, i.e., the appearance of damped oscillations with a certain resonance frequency in response to external synchronous spike volley input (so-called **pulse packets**; Aertsen et al., 1996; Kumar et al., 2010). Subsequently, the investigation shifted to the oscillatory (Osc) regime, characterized by the emergence of spontaneous oscillations in the network activity.

Within each regime, both static and plastic inter-layer connections were considered. We examined three scenarios for the modification of inter-layer connections: (i) FF plastic connections in a FF bi-layer network, (ii) static FF and plastic FB connections in a reciprocally connected network, and (iii) plastic FF and FB connections in a reciprocally connected network.

In each scenario, the initial strengths and delays of FF and FB connections were drawn independently from normal distributions. For plastic connections, the initial weights and delays were determined, based on assigned distributions, with the potential to change via the STDP rule. By contrast, static inter-layer connections retained their weights and delays as per the assigned distributions.

Inter-layer connections were exclusively excitatory. In experiments with static connectivity, the inter-layer connection probability was set to 0.5, involving 70 projection neurons per layer. As a result, each projection neuron in the second layer received 40 excitatory inputs from within its own layer and 35 excitatory inputs from the other layer, on average. For plastic inter-layer connections, the initial connectivity between projection neurons was set to all-to-all in the main simulations. To test the robustness of the results, we also performed control simulations with reduced connection probabilities (75%, 50%, and 25% of the full connectivity). In these cases, synaptic weights were scaled to maintain comparable total synaptic input across conditions.

Self-synapses and multiple synapses were not included in the network configuration. Additional details on the network parameters are given in Table 1.

External input and pulse-packet stimulation

To achieve the AI network state indicated in Figure 1b (left column), each excitatory neuron in each layer was driven by 6,700 independent Poisson excitatory spike trains, each with a mean rate of 1 spike/s. Also, each inhibitory neuron in each layer was driven by 4,000 independent Poisson excitatory spike trains, each with the same mean rate. However, for the Osc state represented in Figure 1b (right column), we changed the input to the excitatory neurons within each layer to 7,500 independent Poisson excitatory spike trains, again each with a mean rate of 1 spike/s. The input to the inhibitory neurons in each layer remained the same. These input parameters, though, were systematically varied in Figure 2a to investigate the state space of possible network activity regimes.

To probe signal propagation and to drive synaptic plasticity in the stimulus-driven conditions, we employed temporally structured external inputs in the form of pulse packets. A pulse packet consisted of a synchronous spike volley delivered to a subset of projection neurons in the first layer network (L1). Specifically, each pulse packet targeted 50 excitatory projection neurons, with spike times drawn from a Gaussian distribution centered at time t_n with a standard deviation of $\sigma = 2$ ms, thereby introducing a small temporal jitter. Pulse packets were delivered at stochastically jittered intervals, with a specified mean inter-packet interval depending on the simulation condition (e.g., 500 ± 100 ms or $1,000 \pm 200$ ms).

Importantly, pulse-packet stimulation was used only in the AI regime, where it served both as a probe of signal transmission between the two layers and, in the case of plastic synapses, as a driver of synaptic modification. By contrast, in the Osc regime, no external pulse packets were applied. Instead, synaptic plasticity was driven entirely by the ongoing endogenous Osc activity of the network, which provides temporally structured spike patterns analogous to repeated input events. This distinction allows us to separately examine delay selection under externally driven and internally generated dynamical conditions.

Data analysis

Signal-to-noise ratio (SNR). To quantify the propagation of input pulse packets, we estimated the SNR based on population spike counts. Specifically, we computed the variance of the spike count signal $y(t)$ in the receiver layer network (L2) following pulse-packet stimulation of the sender layer network (L1) and normalized it by the variance of the same signal during ongoing (unstimulated) activity. The SNR was defined as:

$$\text{SNR} = \frac{\text{Var}[y(t)^{\text{stim}}]}{\text{Var}[y(t)^{\text{ongoing}}]}. \quad (5)$$

Power spectral density (PSD) analysis. The spiking times for each layer network were converted into a spike count vector with a bin size of 1 ms. To analyze the frequency content of spiking activity over time in the Osc network state, we employed Welch's method, a widely used technique for estimating the PSD. This method partitions the spike count data into overlapping windows, each with a 50% overlap between consecutive segments ([0 s, 10 s], [5 s, 15 s],...). The PSD was then computed for each window segment.

Simulation tools

Network simulations were performed using the simulation tool NEST (www.nest-initiative.org; Gewaltig and Diesmann, 2007; Morrison et al., 2007; Linssen, 2018), interfaced with PyNest. The differential equations were integrated using fourth order Runge–Kutta with a time step of 0.1 ms.

In NEST, the delay implemented in connection models corresponds to dendritic delay. To introduce an additional axonal delay, which was fundamental to our study, two intermediate so-called “dummy neuron” populations were included for both forward and backward directions (Fig. 1a). These dummy neuron populations were configured to introduce the desired axonal delay required for the specific simulation. For each inter-layer connection, one dummy neuron was considered, such that any dummy neuron had only one input and one output connection.

Connections from the presynaptic neuron to the dummy neuron were strong, so that each dummy neuron was activated following each spike from the presynaptic neuron connected to it, after a time delay that was the desired axonal delay in the experiment. The connection strength from dummy neurons to postsynaptic neurons was equal to the corresponding biological synapse. Connections from dummy neurons to target neurons were subjected to the plasticity rule. Additionally, dendritic delays were implemented in these connections. In other words, the output connection of each dummy neuron targeted a postsynaptic neuron, included a dendritic delay, and its synaptic strength was equal to the strength of the actual connection between the pre- and postsynaptic neurons. Then, this synapse could be modified through the STDP rule (Fig. 1*a*).

It is worth noting that delays could be honored on both incoming and outgoing connections from and to dummy neurons. We assigned the fixed value of 0.1 ms to the dendritic delays, whereas the axonal delays (those of connections between presynaptic neurons and dummy neurons) were drawn from a normal distribution with mean and standard deviations indicated in each figure caption, accordingly.

By incorporating these intermediate dummy neuron populations, the overall delay in the network could be adjusted appropriately. Note that the rationale behind introducing dummy neurons into the network model was to shift the spike time of the presynaptic neuron by an amount determined mainly by the given axonal delay, while the

weight of its (only) incoming synapse was of no importance, having no biological counterpart.

Results

Baseline activity of the network

Our model consisted of two networks (L1 and L2), each comprising 200 excitatory and 50 inhibitory neurons. Neuronal and synaptic parameters were identical across the two networks and, hence, both layers exhibited the same intrinsic resonance frequency. Within each network, neurons were recurrently connected (intra-layer connection probability = 0.2, independent of neuron type). A subset of 70 excitatory neurons (projection neurons) in each layer projected to the other layer, forming bi-directional connectivity between the networks (Fig. 1*a*).

First, we studied the two networks in isolation, without the projections being present. We tuned the external Poisson input rates to excitatory and inhibitory neurons such that the networks operated in an approximately AI regime under baseline conditions (Fig. 1*b*, left). In this regime, neuronal firing is irregular and only weakly correlated across the population. This is evident from the broad inter-spike interval distributions and the absence

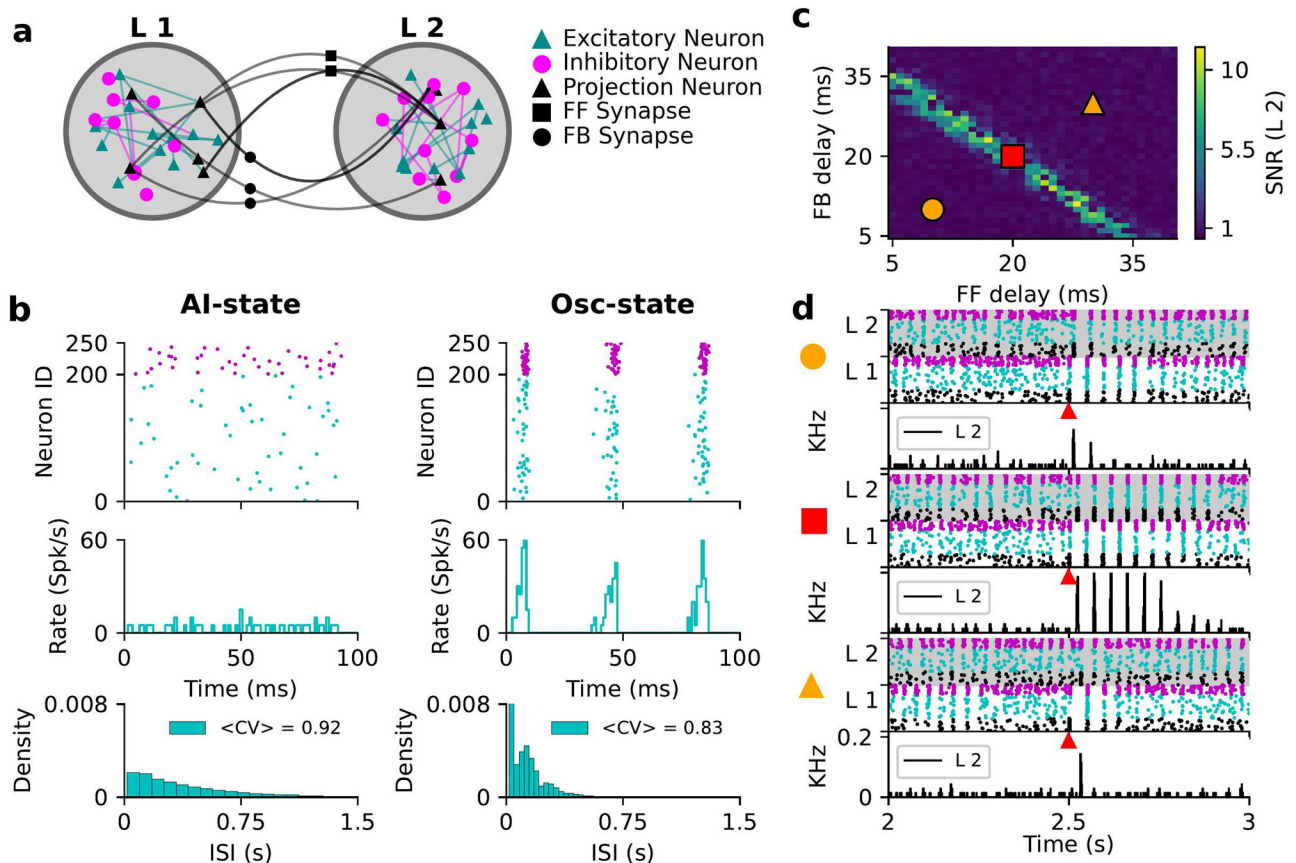


Figure 1. Resonance properties of bi-directionally coupled EI networks **a**, Schematic representation of the canonical circuit consisting of a bi-directionally coupled bi-layer network. Each layer contained excitatory (green triangles) and inhibitory (magenta circles) neurons, sparsely connected by intra-layer static synapses. Projection neurons (black triangles) in the two layer networks were connected via feedforward (FF) and feedback (FB) connections, with inter-layer transmission delays drawn from a normal distribution. FF and FB connections were either static or plastic, depending on the simulation. **b**, Spiking activity of excitatory and inhibitory (E/I) neurons (cyan and magenta; top) and inhibitory synaptic conductance on excitatory neurons (middle) for two input rates producing asynchronous-irregular (AI, left) and oscillatory (Osc, right) dynamics. Input rates were 6,700 spikes/s (left) and 7,500 spikes/s (right). Bottom: inter-spike interval distributions for the excitatory neurons. **c**, FF and FB transmission delays were varied independently to show that matching their total delay to the network's intrinsic resonance period enabled stable pulse packet propagation. Three representative delay combinations are highlighted (orange circle, red square, and orange triangle), corresponding to distinct values of feedforward (FF) and feedback (FB) delays. For each combination, the signal-to-noise ratio (SNR) was computed to quantify the reliability of pulse propagation, where SNR denotes the ratio between stimulus-evoked and baseline activity spike count variance (see Methods). **d**, Spiking activity of two connected layers for the same three FF–FB delay combinations indicated in (**c**). Spiking activity (top) and population activity (bottom) in response to a pulse packet (injected into layer 1 at time $t = 2.5$ ms). The three examples correspond to the three different combinations of FF and FB delays marked in panel **c**. Black, cyan, and magenta indicate projection excitatory, excitatory, and inhibitory neurons, respectively.

of strong population synchrony (Fig. 1*b*, left). For stronger external Poisson input rates, the networks moved into an Osc regime (Fig. 1*b*, right; Rezaei et al., 2020), characterized by increased spiking regularity and synchrony.

In the AI regime, each of the two networks exhibits an intrinsic resonance that becomes apparent in its response to transient inputs. Specifically, following the injection of a synchronous spike volley (a so-called pulse packet; Aertsen et al., 1996; Kumar et al., 2010), the network displays damped Osc activity, the dominant frequency of which defines the resonance frequency of the system (Supplementary Fig. S1). For the chosen parameters, this resonance frequency is approximately 25 Hz. This resonance plays a central role in shaping the interaction between synaptic delays and STDP in the network.

Next, we connected the two networks and systematically varied the FF (from L1 to L2) and FB (from L2 to L1) delays. For each configuration of FF and FB delays we injected a pulse packet into the sender network (L1) and measured the signal propagation. Consistent with our previous work (Rezaei et al., 2020), signal propagation (measured by SNR, see Methods), was maximal when the combined FF and FB delays matched the network's resonance period (Fig. 1*c*).

The intrinsic oscillation frequency of our two networks L1 and L2 was ≈ 25 Hz. Hence, signal propagation was most effective when the sum of FF and FB delays approached 40 ms (Fig. 1*c,d*).

Plastic connections in the FF network

To investigate how inter-layer connections evolve under the assigned STDP rule and to clarify the effect of the network's resonance properties on potentiation and depression of inter-layer connectivity, we connected the two network layers through FF connections. These connections were allowed to change according to the STDP rule. To explore the role of initial delays in determining the polarity (increase or decrease) of synaptic changes, we assumed full connectivity between the excitatory projection neurons in the two populations ($\epsilon = 1$). We chose the initial inter-layer delays from a Gaussian distribution (Fig. 2*a*, blue histograms). We then delivered temporally dispersed pulse packets (number of spikes = 50, sd = 2 ms) at temporally jittered intervals (interval = 500 ± 100 ms). The simulation time was 500 s, but pulse packets were delivered to the projection neurons in the first layer at times between 1 and 490 s (the learning window).

We found that, regardless of the mean of the initial delay distribution, connections with shorter initial delays were potentiated, whereas those with longer initial delays were depotentiated (Fig. 2*a*). After learning, if the synaptic weight fell below a threshold (Fig. 2*b*, horizontal dashed line), we considered it as a pruned connection. The distribution of delays of the surviving connections was negatively skewed, with a mean smaller than the mean of the initial distribution (Fig. 2*a*, compare red and blue distributions in the bottom panels).

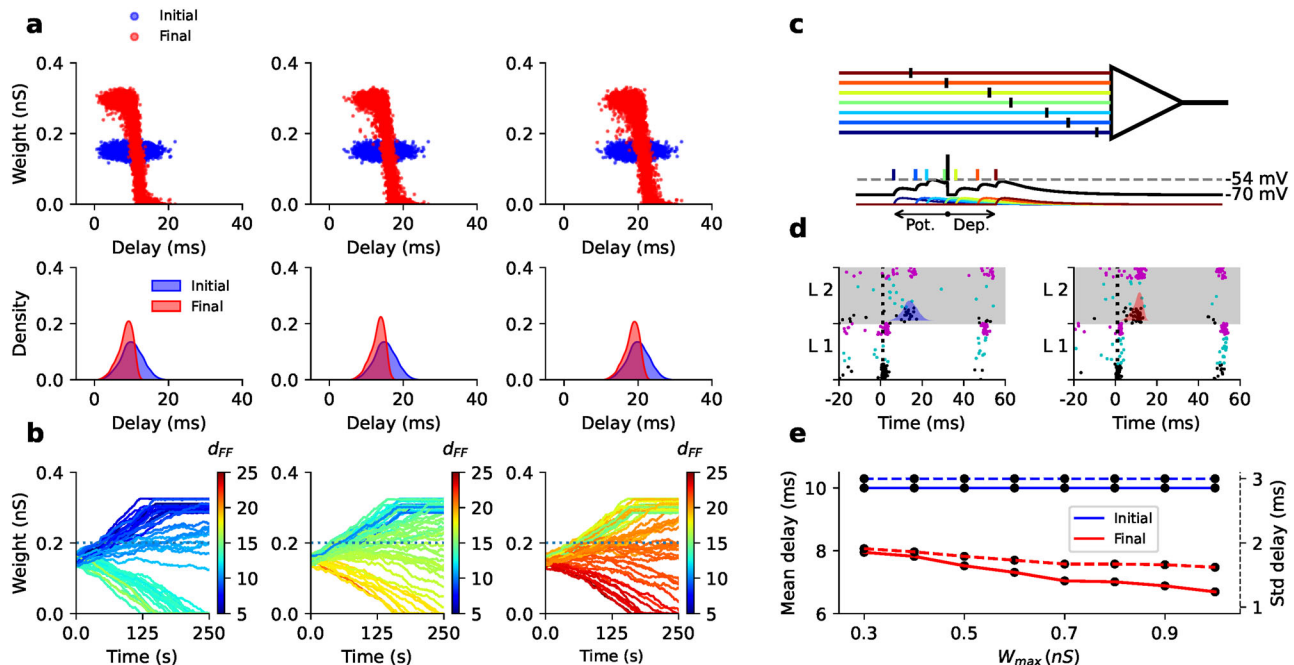


Figure 2. Shorter delays were potentiated in an FF network. *a*, In a bi-layer network with only FF connections, STDP favors connections with shorter delays. Top panels show scatter plots of synaptic delays and weights at initial (blue) and final (red) states; bottom panels show corresponding delay distributions for three initial mean delays (10 ms, 15 ms, 20 ms; $\sigma = 3$ ms). *b*, Evolution of 50 randomly selected FF connections per condition, color-coded by initial delay, showing potentiation of short-delay and depression of long-delay connections. *c*, Schematic of how synaptic delays shape STDP at a single postsynaptic neuron. In the top panel, horizontal lines represent incoming connections with different transmission delays, color-coded such that warmer colors indicate longer delays. These delays determine the relative arrival times of presynaptic spikes at the postsynaptic neuron. In the bottom panels, vertical ticks denote presynaptic and postsynaptic spikes; the color of each presynaptic spike matches the corresponding connection in the top panel. Inputs with shorter delays (cooler colors) arrive earlier and tend to precede the postsynaptic spike (pre-post pairing), leading to synaptic potentiation. By contrast, inputs with longer delays (warmer colors) arrive later, typically following the postsynaptic spike (post-pre pairing), resulting in synaptic depression. *d*, Feedforward (FF) network response before and after learning. After learning, late spikes are reduced due to the depression of long-delay connections and the second layer exhibits stronger and more temporally focused activation. Left and right panels show the initial and final spiking responses, respectively; the vertical dashed line marks the center of mass of the injected pulse packet in layer 1. The initial and final distributions of synaptic delays are overlaid on the raster plot of the second layer spiking response, providing an estimate of the arrival times of incoming spikes before and after learning. Because the network is strictly FF, no inter-layer reverberation or “echo” packets are present and the secondary peaks observed in layer 1 arise from intrinsic resonance within the layer. *e*, With initial FF delays ($\mu = 10$ ms, $\sigma = 3$ ms), increasing W_{max} reduced both the mean (solid) and SD (dashed) of final delays, indicating faster, more reliable transmission. Synaptic weights were drawn from normal distributions: initial $\mu = 0.15$ nS, $\sigma = 0.01$ nS; maximum $\mu = 0.3$ nS, $\sigma = 0.01$ nS; $W_{threshold} = 0.2$ nS. Pulse packets were delivered at intervals 500 ± 100 ms over a 500 s simulation. All panels correspond to a purely feedforward network without feedback projections.

Furthermore, we found that increasing the mean of the initial inter-layer connection delays led to a corresponding increase in the mean of the final delay distribution, though it remained lower than the initial mean in all cases studied (Fig. 2*a*). That is, connections with shorter delay were more likely to survive and to strengthen (as schematically shown in Fig. 2*c*). Thus, under the influence of STDP, synaptic weights were not independent of the synaptic delays (Fig. 2*a*, red dots). Additionally, across all three examples, we observed an increase in the mean of the final synaptic weight distribution, compared to the initial distribution, irrespective of the initial mean delay (Fig. 2*a,b*).

The reason why connections with shorter delays got potentiated and those with longer delays were depotentiated is easy to understand: Each neuron in the L2 network received inputs from L1 via connections with distributed delays. If the initial connectivity is strong, inputs with shorter delays may make the postsynaptic neurons spike before the arrival of inputs from the long-delay connections (Fig. 2*c,d*). Therefore, while the shorter delay connections have pre-post pairing and, hence, get potentiated, long-delay connections effectively end up as post-pre pairing and, hence, get depotentiated.

The potentiation of short-delay projections strengthened the effect of early FF inputs on the second layer, leading to earlier activation of its neurons. As a result, a larger proportion of incoming spikes to the second layer arrived after the postsynaptic spikes, leading to further depression of long-delay connections through the STDP rule. This suggests that increasing the upper limit of synaptic weights causes the depression of more synapses, leading to a narrower final distribution of delays (Fig. 2*e*).

It should be noted that during the propagation of a pulse packet from L1 to L2, the mechanism underlying the selection of short-delay projections depends on the balance between excitation and inhibition, as well as on the recurrent inhibition in the target layer network (L2). The EI-balance causes the postsynaptic neuron to respond preferentially to the earlier inputs, leading to a potentiation. The synchronous inhibitory activity of L2 rapidly follows the synchronous activity of the excitatory projection neurons of L2 (triggered by pulse packet input from L1) and, thus, significantly reduces the effectiveness of the inputs, leading to a depression. Hence, synapses of postsynaptic neurons that primarily received long-delay projections, arriving after inhibition became dominant, were neither depressed nor potentiated.

Evolution of FB connections in the bi-directionally connected network

Next, we bi-directionally connected the two layer networks with an inter-layer connection probability of 0.5. Connection delays were chosen from normal distributions. To characterize the evolution of FB connections, we fixed the FF connections, whereas the FB connections were allowed to change according to the STDP rule. We then injected pulse packets into the L1 network at intervals of $1,000 \pm 200$ ms for 1,990 s. In these simulations, the learning time window was set from 5 to 1,995 s and simulations were run for 2,000 s.

First, we chose the FF delays from a Gaussian distribution (mean = 8 ms, sd = 1 ms, Fig. 3*a*, bottom-left, gray distribution). FF weights were drawn from a Gaussian distribution with mean = 0.2 mS, sd = 0.01 ms. FB connection delays were chosen from a different Gaussian distribution (mean = 25 ms, sd = 8 ms, Fig. 3*a*, bottom-left, blue distribution). The initial strength of each FB connection was also drawn from a Gaussian distribution (mean = 0.2 nS, sd = 0.01 ms, Fig. 3*a*, top left, blue

distribution). The weights and delays of the FB connections were initially uncorrelated.

Given repeated exposure to the pulse packet input, FB connections changed over time (Fig. 3*b*, left). As expected, synaptic changes depended on their respective connections delays. In this example, most potentiated connections lay in a small range of delays (25 – 35 ms). Outside this range, all synapses were depotentiated (Fig. 3*a*, top left, red histogram). Clearly, STDP created a correlation between synaptic weights and delays. Similar to the previous Section, we disregarded all connections with weight smaller than a threshold (0.2, Fig. 3*b*), and observed that the resulting delay distribution had a mean delay of ≈ 30 ms.

Similar results were obtained when we started the simulations with larger mean delays in the FF connections (Fig. 3*a*, bottom, middle, and right panels). However, as we increased the mean delay of the FF connections, the mean delay of the surviving FB connections decreased. In all cases, however, the sum of the maximum delay of the potentiated synapses in the FB direction with the maximum forward delay remained close to the resonance period (Fig. 3*b,c*). In Figure 3*c*, we demonstrate that not only the sum of the maximums of FF and survived FB delays aligns with the network's resonance period (black dotted line), but also that the sum of their mean values shows a close correspondence (red dotted line). Nevertheless, the match is even more precise in the former case.

As expected, the number of surviving (potentiated) FB connections depended on the presence of FF connections with matching delays in the initial distribution. Figure 3*d* shows the number of potentiated FB connections as the mean of the initial FF delay distribution was systematically varied. This result indicates that the number of potentiated FB connections peaked when the sum of the mean FF delay and the mean initial FB delay matched the resonance period of the two layer networks.

Applying the STDP rule to FB connections enabled the bi-layer network to approach resonance. Initially, the broad range of delays in the FB pathway caused backward-propagating spikes to be temporally dispersed. However, after STDP induced a selection of suitable FB connection delays, spikes from network L2 arrived in network L1 much more synchronously (Fig. 3*e*).

Self-organization of the delays when both FF and FB connections were plastic

We next made both FF and FB connections plastic. In the initial state, both FF and FB connection weights and delays were drawn from a Gaussian distribution, with the initial FF delay distribution being narrower than the initial FB delay distribution (Fig. 4*a*, left). Note that as in the previous simulations, initial synapse weights and delays were not correlated. To train the FF and FB connections, we injected pulse packets into L1 at intervals of $1,000 \pm 200$ ms. In these simulations, pulse packets were injected only between 5 and 2,495 s (learning time window) and the networks were simulated for a total of 2,500 s.

As in the previous simulations, STDP potentiated synapses with specific delays and, thereby, created a correlation between synaptic weights and delays for both the FF and FB connections (Fig. 4*b*, left). We found that FF connections consistently maintained shorter delays in the final distribution. This indicates that, regardless of the networks' resonance properties, FF connections with longer delays were pruned during the transmission of pulse packets in the forward direction (Fig. 4*a*).

By contrast, FB connections were tuned according to the resonance properties of the individual networks. A change in the mean of the FF delay distribution also resulted in a

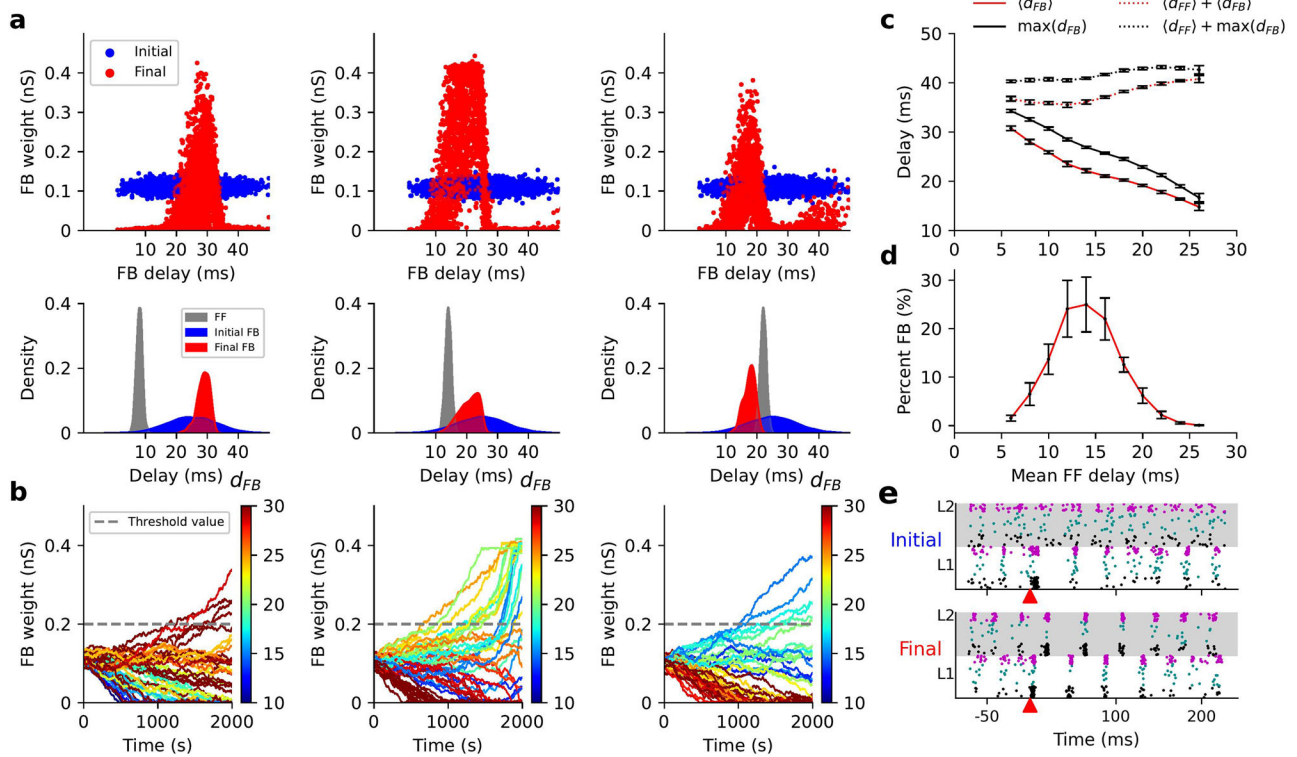


Figure 3. Delay selection of FB connections based on resonance with static FF connections. **a**, Top panels show the raster plots of the delays and weights of the FB connections at initial (blue) and final (red) states. Bottom panels show the corresponding distribution of the initial delays and final delays of the potentiated FB connections. The FF connections were chosen to be static, with a delay distribution of $\sigma = 1$ ms, and three different mean values (8, 14, and 22 ms), as shown in gray in the bottom panels. The FB connections evolved through the STDP rule, initialized with a normal delay distribution ($\mu = 25$ ms, $\sigma = 8$ ms). As the mean delay of the FF connections increased, the mean of the final distribution of FB delays decreased, resulting in a peak at lower values. **b**, Evolution of weights for the FB connections with different delays is shown for each set of initial parameters in (**a**). Legends show the color code used to indicate different delays. **c**, The mean of the final distribution of FB delays decreased as the mean FF delays systematically increased (red curve). The FB connections evolved such that the final mean delays, summed with the mean FF delays, remained slightly below the resonance period of both networks. The error bars indicate the standard deviation over the 20 trials. The sum of the maximum values of the final delays in both directions, almost matched the resonance period. **d**, The number of potentiated FB connections, for different values of the mean delay of FF connections. **e**, Spiking activity of the two layers at initial and final states. A pulse packet was injected into the first layer network at time 0 marked by red triangles. FF and FB weights were drawn from normal distributions with parameters ($\mu = 0.2$ nS, $\sigma = 0.01$ nS) and ($\mu = 0.11$ nS, $\sigma = 0.01$ nS), respectively. The maximum weights allowed also followed a normal distribution ($\mu = 0.4$ nS, $\sigma = 0.01$ nS). Additionally, $W_{\text{threshold}}$ was set to 0.2 nS. Pulse packets were injected into the L1 network at intervals of 1 s with ± 200 ms jitter during the period from 5 to 1,995 s, for a total simulation time of 2,000 s.

corresponding change in the FB delay distribution: increasing the mean FF delay resulted in a decrease in the mean final FB delay (Fig. 4a,c). Across all conditions, the sum of the mean (as well as the sum of the maxima of) final delays for FF and FB connections remained relatively close to the intrinsic oscillation period (respectively; Fig. 4c, red and black dotted curves).

Importantly, the emergence of FB connections with delays that matched the networks' resonance properties depended on the presence of a sufficient number of connections with appropriate delays in the initial FB delay distribution. The mean of the FF delays determined the range of delays that were effectively potentiated through plasticity. Therefore, the number of FB connections that were potentiated depended on the mean FF initial delay (Fig. 4d).

Note that compared to the case of static FF synapses (Fig. 3d), the peak was moved to even shorter FF delays, since here the mean final FF delay itself was shorter than the initial one.

Because FB connection weights and delays depend on the FF delay distribution to favor appropriate delays, FB connections reached their steady state later than the FF connections did (Fig. 4b).

Tuning of FF and FB synapses using STDP led to an improved propagation of signals from L1 to L2. Figure 5 shows that the pulse packets initially failed to reverberate in the network,

whereas after synaptic modification via STDP, successful reverberation between the two layer networks was established. It is worth noting that the facilitation of signal propagation, quantified by an increase in the SNR, did not trivially come with an increase in synaptic cost, defined as the sum of all synaptic weights of FF and FB connections (Fig. 5). While the final synaptic cost depended on the initial distribution of delays and weights of the inter-layer connections, our results suggest that improved signal transmission is not merely a trivial outcome of increased mean connectivity weight between the two layers.

Robustness of delay selection to network parameters

To assess the generality of the delay-selection mechanism, we examined the robustness of our results to two key modeling assumptions: (i) the impact of introducing a mismatch in the intrinsic oscillation frequencies of the two network layers and (ii) the effect of reducing the initial inter-layer connectivity. In all cases, network parameters were adjusted to isolate the effect of interest, while preserving overall synaptic input and network dynamics.

The results presented thus far were obtained for two networks with matched intrinsic oscillation frequencies. To evaluate the robustness of the delay-selection mechanism under more heterogeneous conditions, we introduced a mismatch between the

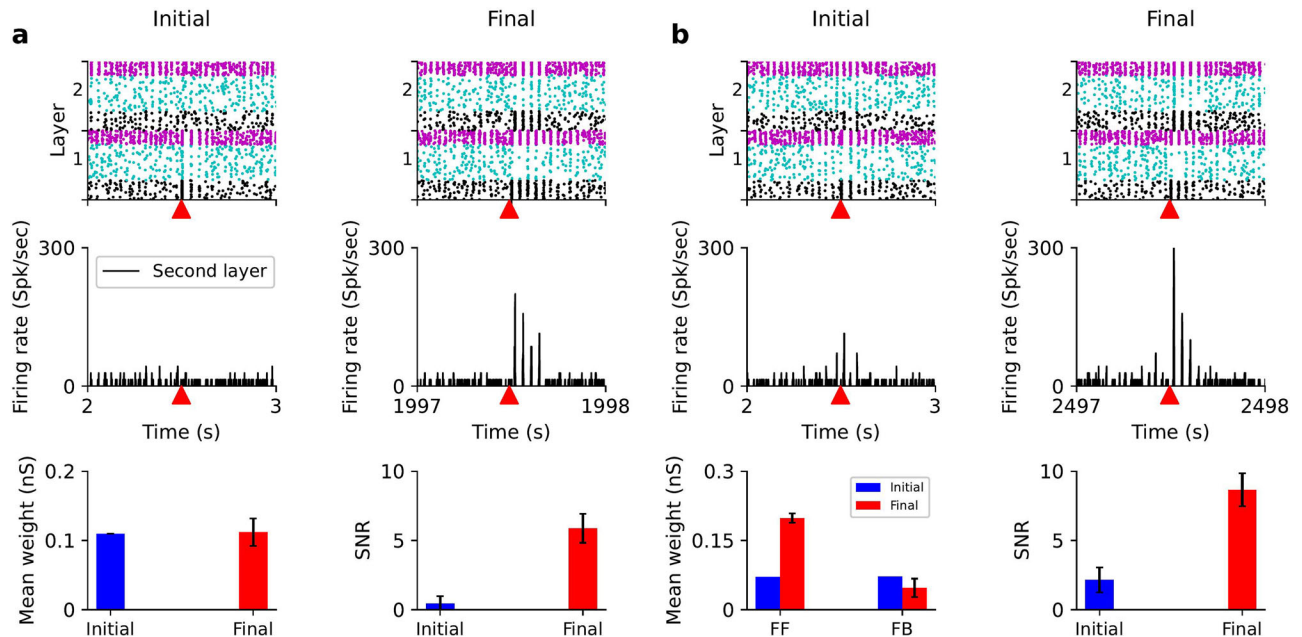
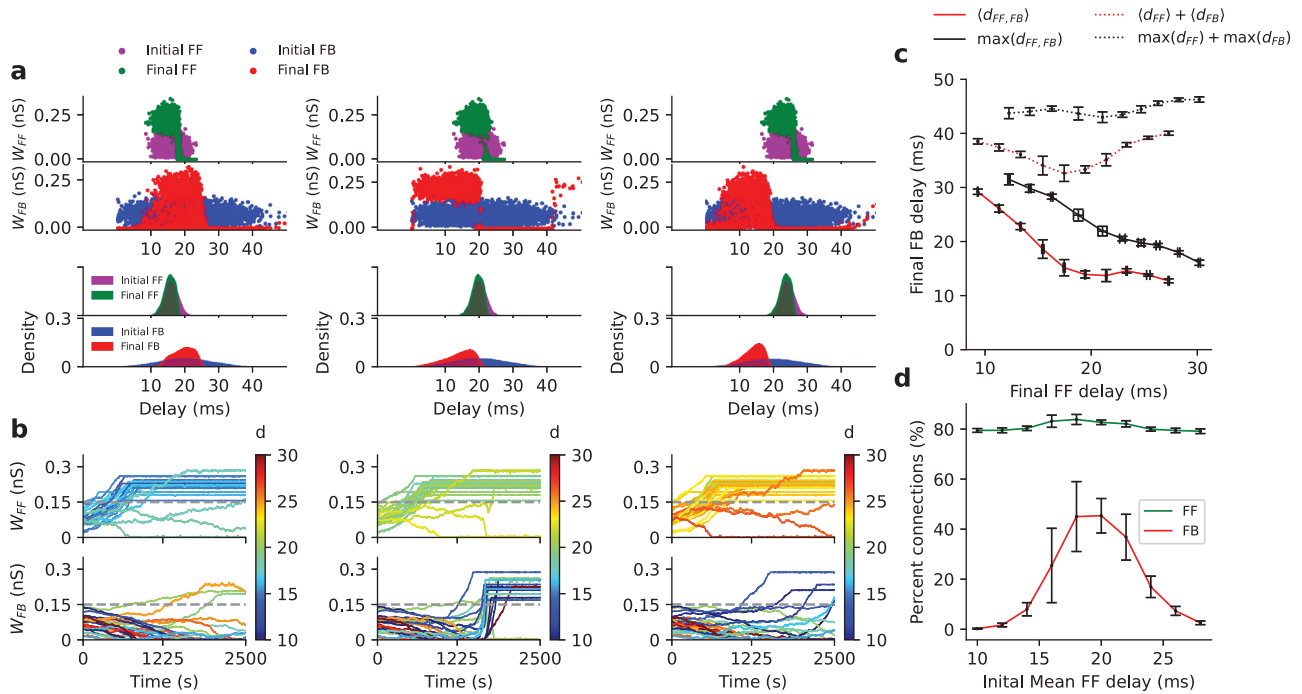


Figure 5. STDP facilitates pulse packet propagation through delay selection. A bi-layer network successfully transmitted a single pulse packet ($\alpha = 40$, $\sigma = 2$ ms) after modification of connectivity under two distinct configurations: static FF and plastic FB connections **a**, and both plastic FF and FB connections **b**. In both configurations, the second layer exhibited a high SNR due to the alignment of inter-layer delays with the resonance period of the individual layer networks. By stimulating the first layer with a single pulse packet, we recorded the spiking response in the second layer. The time of pulse-packet stimulation is indicated by red triangles. After synaptic modification through STDP, the initial pulse packet gives rise to subsequent activity peaks, reflecting reverberatory propagation between the two layers rather than repeated external inputs. The stronger reverberation and higher SNR observed after learning do not rely on excessive synaptic strengths, as the mean weight of the resonant connections remains around or even below their initial values (bottom panels). The parameters used in **(a)** and **(b)** were the same as those in Figures 3 and 4, respectively and, therefore, the initial mean FB weights differ between the two panels. The error bars indicate the standard deviation over 20 trials.

intrinsic frequencies of the two layer networks. The resonance frequency of a network can be altered by various neuron parameters and connection strengths (Hahn et al., 2014). Here we varied the inhibitory synaptic time constant in the second layer network to shift its resonance frequency, while keeping the rest of the network parameters unchanged (Fig. S1).

We found that for moderate frequency mismatches (± 2 Hz), the two networks remained capable of phase-locking following pulse stimulation, and the delay-selection mechanism persisted (Fig. S2). In this regime, the selection of shorter FF connections remained robust and largely independent of the resonance frequency of the second network. The FB connections that survived were also shaped by the resonance properties of the coupled system: FF and FB delays adjusted such that their combined delay aligned with the effective period of the joint dynamics. When the resonance frequency of the second layer network was much smaller than that of the first layer network, the delay-selection effect remained, however, the number of connections that survived and potentiated decreased. In particular, for larger mismatches, the proportion of FB connections that underwent potentiation was reduced, eventually approaching zero at the largest mismatch (Fig. S3). This suggests that when there is a larger mismatch in the resonance frequency of the two networks, their interaction becomes largely FF.

The all-to-all connectivity used in previous simulations allows for maximal competition between synapses with different delays. However, this choice could in principle bias STDP toward favoring earlier-arriving inputs. To test whether the observed delay-selection mechanism depends on the initial inter-layer connectivity, we repeated the simulations using sparser initial connectivity.

Specifically, we reduced the inter-layer connection probability to 75%, 50%, and 25% of full connectivity. To ensure that any observed differences were not trivially due to reduced input strength, synaptic weights were rescaled in each condition to preserve the total synaptic input to the postsynaptic neurons. As shown in [Supplementary Figures S4 and S5](#), the qualitative trends remained consistent across all tested conditions. In particular, FF connections continued to favor shorter delays, whereas FB connections were shaped by the resonance properties of the network. The distributions of potentiated FF and FB delays remained consistent with the resonance condition, such that their combined delay stayed close to the intrinsic oscillation period of the network.

Together, these results demonstrate that the emergence of delay selection through STDP is not restricted to highly specific parameter choices, but instead reflects a robust interaction between synaptic plasticity and Osc network dynamics.

Delay selection in the Osc regime

Previous theoretical work has shown that synchronization in delay-coupled Osc networks depends on conduction delays, and that synaptic plasticity can shape coupling in such systems (Ermentrout and Kopell, 1998; Karbowski and Ermentrout, 2002; Sadeghi and Valizadeh, 2014; Madadi Asl et al., 2017). Thus far, we have presented pulse packets to the first layer network to train the synapses. We then hypothesized that spontaneous oscillations in the dynamics of neuronal networks can facilitate the formation of the connections in a specific range of delays. That is, Osc activity can provide pulse-packet-like input to the network and train the synapses. To test this hypothesis, we set the parameters of the two networks such that they

operated in an Osc regime (Osc state SR; Fig. 1*b*, right column) and the population activity showed oscillations at ≈ 25 Hz.

Initially, we considered uni-directional FF connectivity between two identical layer networks, characterized by a delay distribution with an initial standard deviation of 3 ms, and let them evolve through the assigned STDP rule. We found that, regardless of the initial mean delay, connections with shorter delays were preferentially potentiated (Fig. 6*a*, left panel). As a result, both the mean and the standard deviation of the delay distribution decreased in the final state. Furthermore, increasing $W_{\max}$ led to a progressive reduction in both the mean and the width of the final delay distribution (Fig. 6*a*, right panel), resembling the behavior observed in the case of evoked activity discussed earlier (Fig. 2*d*). Notably, these simulations predict that, in spontaneously Osc networks, the Osc activity of the L2 population would become phase-locked to that of the L1 population, with a time lag approximately equal to the mean delay of the inter-layer connections. Consistent with this hypothesis, the time lag between L1 and L2 population activity scaled with the mean of the initial delay distribution (Fig. 6*b*, right panel). Three examples of phase-locked spiking activity of the two layer networks are represented for the three different values of FF delays (Fig. 6*b*, left panel).

We next considered the reciprocally connected networks. To break the network's symmetry and identify up- and down-stream layers, we initialized the FF connection delays using a relatively narrow distribution. By contrast, FB connections were initialized with a wider delay distribution, consistent with the case of evoked activity discussed earlier (Fig. 4).

We observed that the evolution of the connections ensured that the sum of the final mean delays in both directions remained approximately constant, aligning with the period of the Osc activity (Fig. 6*c*). Furthermore, the means of the final delay distributions determined the phase relationship between the oscillations of the bi-layer networks. Consistent with the emergent delay distributions, the phase difference between the oscillations of the two layer networks corresponded to the delays in both directions. As shown in [Figure 6*d*](#) (left panel), the time difference in spiking activity between the first and second layer networks aligned closely with the mean of the delay distribution in each configuration.

A key observation regarding the evolution of inter-layer connectivity in the spontaneous Osc state is that the formation and stabilization of consistent delays occurred significantly faster than in the stimulus-evoked activity in the AI state studied earlier (Fig. 6*f*; compare to Fig. 4). This difference is reasonable since, in the stimulus-evoked activity in the AI state, effective weight modification happened only upon the impact of each input pulse packet. By contrast, in the Osc regime, weights were updated at each oscillation cycle, which is much shorter than the intervals between consecutive pulse packet inputs in the stimulus-evoked AI state, which was set to be around 1 s.

Discussion

The adaptability of signal transmission speed and synaptic delays in the brain has been observed in experimental studies, suggesting that delays are not static, but can change dynamically in response to neuronal activity (Katz and Miledi, 1965; Stanford, 1987; Boudkkazi et al., 2007; Bakkum et al., 2008; Boudkkazi et al., 2011; Boudkkazi and Debanne, 2024). However, the specific rules governing this delay plasticity remain largely unknown. Computational studies have proposed mechanisms in which delay plasticity operates in parallel with synaptic weight

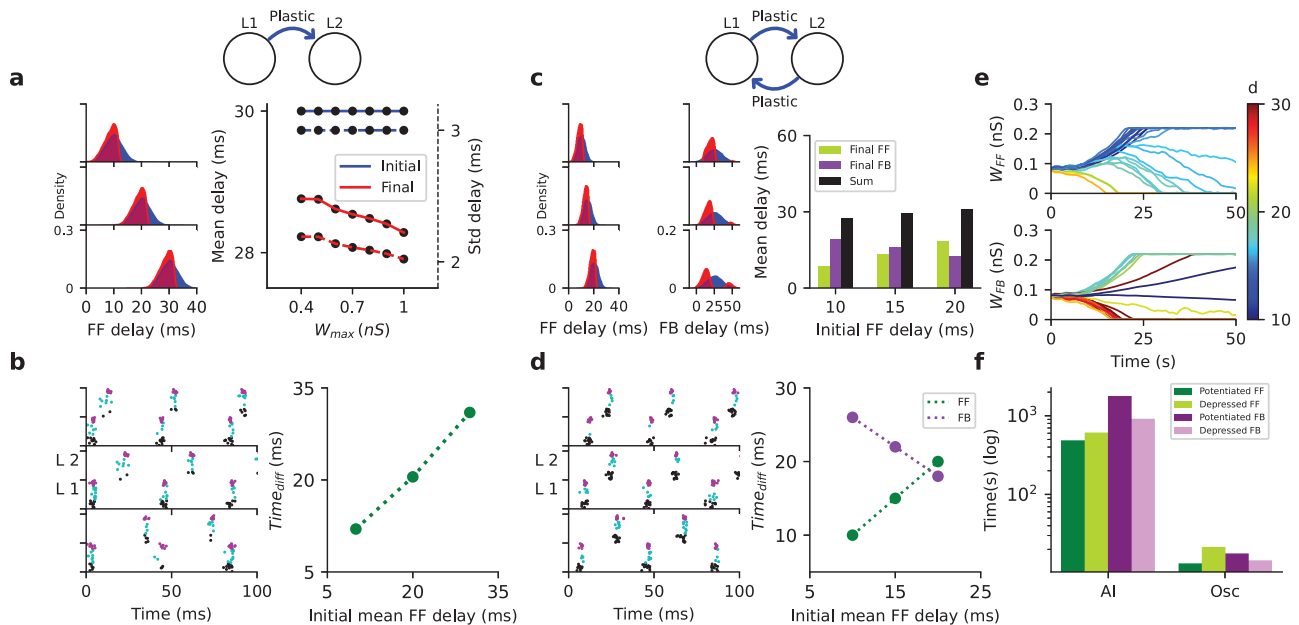


Figure 6. Delay selection in uni-directional and bi-directional oscillatory (Osc) layer networks. **a**, In a bi-layer Osc network with uni-directional projections, STDP favored short-delay connections for all initial mean delays. Left: initial (blue) and final (red) inter-layer delay distributions after 30 s of learning. Right: increasing $W_{\max}$ reduced the mean and SD of the final delay distribution. Solid curves indicate the mean of the distributions, while dashed curves denote the corresponding standard deviations (SD). Initial mean FF weight was 0.2 nS; $W_{\max} = 0.4$ nS, $W_{\text{threshold}} = 0.2$ nS. **b**, Spiking activity for each initial condition during 100 ms after 26 s of learning. Right: relation between inter-layer spike-timing difference and initial mean FF delay. Spike times were computed as the center of mass of synchronized volleys, averaged across phase-locked events within 1 s after 26 s of learning. **c**, In bi-directionally connected Osc networks, both FF and FB connections evolved via STDP. For three initial mean FF delays ($\mu = 10, 15, 20$ ms, $\sigma = 3$ ms), the final delay distributions after 50 s (left; blue: initial, red: final) showed that total loop delays (black bars) remained constant (right). FB delays had $\mu = 25$ ms, $\sigma = 10$ ms. Initial inter-layer weights: 0.08 nS; $W_{\max} = 0.22$ nS; $W_{\text{threshold}} = 0.15$ nS. **d**, Spiking activity during 100 ms after 46 s of learning (left) and corresponding inter-layer timing differences (right), as in (**b**). **e**, Evolution of 40 randomly selected FF (top) and FB (bottom) connections for initial mean FF delay of 15 ms. **f**, Connectivity over time evolution—both potentiation and depression—occurred faster in the Osc state (panel (**e**)) than in the stimulus-evoked AI state (Fig. 4*b*). Bars show the time for each connection type to cross the weight threshold after potentiation (0.15 nS) or depression (0.03 nS).

plasticity, demonstrating that such dual plasticity can enhance the computational capabilities of neuronal networks (Wang et al., 2015; Yu et al., 2022). By contrast, in our study, synaptic delays are not plastic variables. Instead, we show that structured delay distributions can emerge without invoking an explicit mechanism for delay plasticity. The STDP rule, acting on fixed delays, selectively reinforces connections whose transmission delays are compatible with the temporal structure of network activity.

By allowing a STDP rule to shape synaptic connections based on precise spike-timing, delays naturally evolve to align with the networks' resonance properties, optimizing communication efficiency. This approach builds on earlier work (Gerstner et al., 1996; Izhikevich, 2006), which showed how STDP could refine temporal relationships between neurons' spiking activities. Our findings highlight an additional dimension, though: the ability of STDP to exploit network Osc dynamics to fine tune delays in reciprocally interconnected neuronal networks. This integration of resonance properties into the delay selection process adds another aspect of functional significance, demonstrating how a single plasticity mechanism can simultaneously coordinate synaptic weights and delays to achieve optimal signal propagation and network synchronization.

We speculated that these findings could be of importance for early developmental processes, during which neural circuits undergo extensive refinement driven by spontaneous and evoked activity (Katz and Shatz, 1996). Previous research suggested that early endogenously generated patterns of activity play a crucial role in guiding the formation of neural connections in sensory and motor areas (Hubel and Wiesel, 1959; LeVay and Stryker,

1979; Rao and Ballard, 1999; Hensch, 2005). Our results support this hypothesis by showing that synaptic reorganization induced by spontaneous Osc activity was compatible with that of stimulus-evoked activity, whereas the modification through the endogenous rhythmic activity was considerably faster than that induced by the evoked activity. The continuous nature of spontaneous oscillations enabled rapid weight updates within each oscillation cycle, aligning delays more rapidly and efficiently. This finding underscored the developmental importance of Osc rhythms in shaping the temporal structure of neuronal networks along with the sensory-driven activity.

Moreover, our results suggest that oscillations can stabilize the tuning of delays beyond the brain development period. Disruptions in Osc dynamics and timing mechanisms have been implicated in several neurodevelopmental disorders, including autism, schizophrenia, and epilepsy (Uhlhaas and Singer, 2010). Our findings suggest that aberrant Osc activity during development and beyond could impede the proper tuning of inter-network delays, leading to inefficient communication and cognitive deficits. Investigating how Osc disruptions affect delay tuning in pathological states could provide important new insights into the etiology of these conditions and suggest novel therapeutic strategies.

Connections between layered networks should be flexible enough, so that they can be modified in a context/state dependent manner. Here we have described a strategy by which connections with certain delays were selected to facilitate communication between two layer networks. How could we also unlearn these connections? The simplest way to weaken the connections in the model would be to alter the resonance properties of the

receiver network, which would entail a change in the excitation–inhibition balance and the gain of individual neurons (Gerstner, 1995). In addition, a change in the spiking mode of neurons, for example a change from single spike emission to bursting behavior, could also alter the Osc dynamics and, thereby, weaken the synapses (Sahasranamam et al., 2016).

In general, connectivity between two networks operating in an Osc regime depends on both frequency and phase of the oscillations (Devalle and Roxin, 2025). In fact, interaction between STDP and oscillations can make the network structure bi-stable and, depending on the initial conditions, there can be no connectivity, uni-directional connectivity or bi-directional connectivity. In our model, delays introduced phase differences between the two network oscillations and, therefore, connections corresponding to specific delays survived. However, if some process (e.g., inputs from external sources) were to change the phase of the oscillations, the network connectivity could be completely reorganized (Devalle and Roxin, 2025).

Model limitations and future directions

While our study provided valuable new insights, it simplified certain aspects of cortical circuits. We used a conventional linear profile of STDP (Bi and Poo, 1998; Kistler and van Hemmen, 2000; Wang et al., 2005), to highlight its role in the selection of proper connection delays and synaptic weights. Several theoretical and experimental studies have shown the shortcomings of this type of STDP rule, and have suggested variants and complementary mechanisms (Bienenstock et al., 1982; Song et al., 2000; Van Rossum et al., 2000; Rubin et al., 2001; Froemke and Dan, 2002; Clopath et al., 2010; Yu et al., 2022; Agnes and Vogels, 2024). Moreover, we assumed homogeneity in STDP rules, whereas biological networks exhibit diverse forms of plasticity that may influence delay tuning (Feldman, 2012). Incorporating heterogeneous plasticity rules could refine our understanding of how delays are tuned across different brain areas and Osc regimes.

The model networks we studied consisted of several hundreds of neurons. The question arises, whether the phenomena of delay tuning via STDP and its interactions with network dynamics also hold for much larger networks, as we have them in real brains. As we have shown, the key observation of delay tuning is a property of network oscillations. Therefore, as long as a network is capable of network oscillation and resonance properties, we should observe this kind of delay tuning, independent of the network size, provided the scaling of the various parameters is suitably taken into account. Moreover, both small and larger networks can show oscillations, as we know from many studies in the experimental and theoretical literature.

In the present manuscript, inter-layer projections were restricted to excitatory-to-excitatory connections in order to isolate the role of delay selection under STDP in the simplest reciprocal setting. Adding inter-layer E→I projections would be biologically relevant and could alter both the effective timing competition and the resonance properties of the coupled networks. Meanwhile, the key requirement for the delay selection is that the pulse-packet travels from layer 1 to 2 (and back). E→I projections can affect the transmission of the pulse packet (Vogels and Abbott, 2009; Kremkow et al., 2010), so as long as FF inhibition (due to E→I projections) is lagged with respect to the FF excitation, we expect that the results would remain unaffected by E→I projections.

In addition, our results indicate that the effectiveness of the delay-selection mechanism depends on the ability of the coupled networks to maintain phase-locking. While moderate intrinsic

frequency mismatches can be compensated by adaptive adjustments in the effective loop delay, larger mismatches disrupt phase coherence, weakening the temporal structure required for consistent spike-timing relationships. As a result, delay selection becomes less precise and can eventually break down. This suggests that the emergence of functionally relevant delays is constrained by the dynamical compatibility of the interacting networks.

Another limitation is the assumption of static resonance properties within the coupled networks. In biological systems, resonance frequencies are dynamic and can be modulated by neuromodulators, learning, or disease states (Wang, 2010). Exploring how such modulation influences delay tuning would deepen our understanding of the adaptability of neural circuits.

An important open question concerns the persistence of the learned connectivity and the possibility of unlearning. In the present study, we did not systematically analyze the stability of the learned configuration after the removal of pulse-packet inputs, nor the time scales over which such configurations may persist under altered activity regimes. However, the structure of the STDP rule provides some qualitative insight. In the AI regime, where spike-timing is weakly structured, the negative integral of the STDP kernel is expected to gradually weaken previously potentiated connections in the absence of temporally precise inputs, leading to a progressive loss of the learned configuration (Gerstner et al., 1996; Bi and Poo, 1998; Song et al., 2000). By contrast, ongoing Osc activity can sustain or even reinforce the selected delays by preserving consistent spike-timing relationships across cycles (Kerr et al., 2013; Madadi Asl et al., 2018). This suggests that the long-term maintenance of learned inter-areal connectivity may depend on intermittent or sustained periods of coherent activity, whereas changes in network dynamics—such as shifts in Osc state or resonance properties—could promote the reorganization or decay of previously established pathways. A systematic investigation of these processes, including the conditions under which learned connectivity is stabilized or forgotten, remains an important direction for future work.

Data Availability

The corresponding codes will be deposited in a Github repository and made available upon publication.

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